

MSc Mathematics

Semester 1 — Complete Study Guide

Manipal University Jaipur — Distance Education

Prepared for CSIR NET Qualification & Assistant Professorship

6 Subjects · 60 Units · 35+ Complete Proofs · 120+ Worked Examples

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Chapter 1

Advanced Linear Algebra

1.1 Unit 1: Linear Transformations

Prerequisites — What You Need to Know

Matrix: A rectangular grid of numbers arranged in rows and columns.

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \quad (\text{a } 2 \times 2 \text{ matrix})$$

Matrix multiplication: Row $\times$ Column. Multiply corresponding entries and add:

$$\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \times 1 + 3 \times 0 \\ 1 \times 1 + 4 \times 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Vector: An ordered list of numbers, like $\mathbf{v} = (3, -1, 2) \in \mathbb{R}^3$.

Linear combination: $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ means scaling vectors and adding them.

Definition 1.1.1: Linear Transformation

Let V and W be vector spaces over a field F . A function $T : V \rightarrow W$ is a **linear transformation** if for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars $c \in F$:

- (i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (preserves addition)
- (ii) $T(c\mathbf{v}) = cT(\mathbf{v})$ (preserves scalar multiplication)

In plain English: A linear transformation is a function between vector spaces that “plays nice” with addition and scaling. If you double the input, the output doubles.

Worked Example — Rotation in $\mathbb{R}^2$

The rotation by angle θ counterclockwise is $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by:

$$T(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$$

Step 1: Check additivity. Let $\mathbf{u} = (x_1, y_1)$ and $\mathbf{v} = (x_2, y_2)$:

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T(x_1 + x_2, y_1 + y_2) \\ &= ((x_1 + x_2) \cos \theta - (y_1 + y_2) \sin \theta, (x_1 + x_2) \sin \theta + (y_1 + y_2) \cos \theta) \\ &= \underbrace{(x_1 \cos \theta - y_1 \sin \theta, x_1 \sin \theta + y_1 \cos \theta)}_{T(\mathbf{u})} + \underbrace{(x_2 \cos \theta - y_2 \sin \theta, x_2 \sin \theta + y_2 \cos \theta)}_{T(\mathbf{v})} \\ &= T(\mathbf{u}) + T(\mathbf{v}) \quad \checkmark \end{aligned}$$

Step 2: Check homogeneity: $T(c\mathbf{u}) = cT(\mathbf{u})$ follows similarly. $\checkmark$

Conclusion: Rotation is a linear transformation.

Worked Example — Differentiation as a Linear Transformation

Let $D : P_n(\mathbb{R}) \rightarrow P_{n-1}(\mathbb{R})$ be the derivative operator.

$$D(3x^2 + 2x + 1) = 6x + 2$$

Why is D linear?

- $D(f + g) = (f + g)' = f' + g' = D(f) + D(g) \quad \checkmark$
- $D(cf) = (cf)' = cf' = cD(f) \quad \checkmark$

Definition 1.1.2: Kernel and Image

For a linear transformation $T : V \rightarrow W$:

- **Kernel** (null space): $\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$
Think of it as: “Which inputs does T squash to zero?”
- **Image** (range): $\text{Im}(T) = \{T(\mathbf{v}) : \mathbf{v} \in V\}$
Think of it as: “What outputs can T actually produce?”

Theorem 1.1.1: Rank–Nullity Theorem

If $T : V \rightarrow W$ is a linear transformation and V is finite-dimensional, then:

$$\dim(V) = \dim(\ker T) + \dim(\text{Im } T)$$

Equivalently: $\dim(V) = \text{nullity}(T) + \text{rank}(T)$

Proof. **Step 1:** Let $\dim(\ker T) = k$. Choose a basis $\{v_1, \dots, v_k\}$ for $\ker T$.

Step 2: Extend this to a basis $\{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ for V (where $n = \dim V$).

Step 3: We *claim* that $\{T(v_{k+1}), \dots, T(v_n)\}$ is a basis for $\text{Im}(T)$.

Spanning: Any $w \in \text{Im}(T)$ has the form:

$$w = T\left(\sum_{i=1}^n c_i v_i\right) = \sum_{i=1}^n c_i T(v_i) = \underbrace{\sum_{i=1}^k c_i T(v_i)}_{=0 \text{ (since } v_i \in \ker T)} + \sum_{i=k+1}^n c_i T(v_i)$$

So w is a linear combination of $T(v_{k+1}), \dots, T(v_n)$. ✓

Linear independence: Suppose $\sum_{i=k+1}^n c_i T(v_i) = \mathbf{0}$.

Then $T\left(\sum_{i=k+1}^n c_i v_i\right) = \mathbf{0}$, so $\sum_{i=k+1}^n c_i v_i \in \ker T$.

Therefore $\sum_{i=k+1}^n c_i v_i = \sum_{j=1}^k a_j v_j$ for some scalars a_j .

Since $\{v_1, \dots, v_n\}$ is a basis (linearly independent), *all* coefficients are 0.

In particular, $c_{k+1} = \dots = c_n = 0$. ✓

Conclusion: $\dim(\text{Im } T) = n - k$, so $n = k + (n - k)$. ■

Important — Exam Alert!

The Rank–Nullity theorem is asked in **every** university exam and appears in CSIR NET regularly. You must be able to:

1. State the theorem with correct notation
2. Reproduce the complete proof
3. Apply it: given $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ with nullity = 2, find rank: $5 = 2 + \text{rank} \Rightarrow \text{rank} = 3$

Worked Example — Finding Kernel and Image

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x_1, x_2, x_3) = (x_1 + x_2, x_2 + x_3)$.

Step 1: Find $\ker(T)$. Set $T(x_1, x_2, x_3) = (0, 0)$:

$$\begin{cases} x_1 + x_2 = 0 \\ x_2 + x_3 = 0 \end{cases} \implies x_1 = -x_2, \quad x_3 = -x_2$$

So $\ker(T) = \{t(-1, 1, -1) : t \in \mathbb{R}\}$, and $\dim(\ker T) = 1$.

Step 2: Find rank using Rank–Nullity.

$$\dim(\mathbb{R}^3) = \dim(\ker T) + \text{rank}(T) \implies 3 = 1 + \text{rank}(T) \implies \text{rank}(T) = 2$$

Step 3: Since $\text{rank}(T) = 2 = \dim(\mathbb{R}^2)$, T is **surjective** (onto). ✓

